

Emergency First Aid Training & Real Time Chat Consultancy

A client-side web application designed to teach common first-aid procedures and simulate interactive guidance during emergencies.





Introduction: Empowering Laypersons

1

Learning Modules

Comprehensive first-aid learning content.

2

Emergency Chat

JavaScript-based assistant for real-time guidance.

3

Content Feedback

Feedback section for accurate content display.

4

Browser Support

Runs entirely in the browser with no server dependencies.

The Problem: Bridging the Knowledge Gap

Most people lack first-aid training, leading to hesitation or incorrect actions during emergencies.

Bystander action significantly improves outcomes in critical events like cardiac arrest and choking, yet public knowledge remains low.

Existing mobile and web apps often require backend servers and can only serve for calling emergency services, limiting their utility in critical moments.





Our Solution: Client-Side, Browser-Only Guidance



Browser-Only

No backend server dependencies.



Guideline-Based

Equips users with quick, evidence-based directions.

Literature Review: Impact of Training

OHCA Outcomes

Early bystander CPR improves out-of-hospital cardiac arrest outcomes.

Smartphone-aided training increases CPR rates.

Decision Support

Rule-based web logic offers structured support without AI accuracy concerns.

1

2

3

Effectiveness

Mobile and web apps enhance knowledge retention and preparedness, boosting confidence in procedures.



Filling the Research Gap

While many first-aid tools exist, few offer a comprehensive, client-side solution.

- **Browser-Only Operation**

No backend or server dependencies.

- **Interactive Decision Flows**

Purely HTML/CSS/JS-driven guidance.

- **Simulated Assistance**

No external LLM or AI services required.



Key Objectives

1

Client-Side Web App

Design and implement a purely client-side first-aid training web app.

2

Interactive Flows

Develop decision flows for CPR, choking, bleeding, drowning, allergic reactions, poisoning and burns.

3

Low Data

Ensure minimal data dependency.

4

Intuitive UX

Provide a simple, responsive user experience.

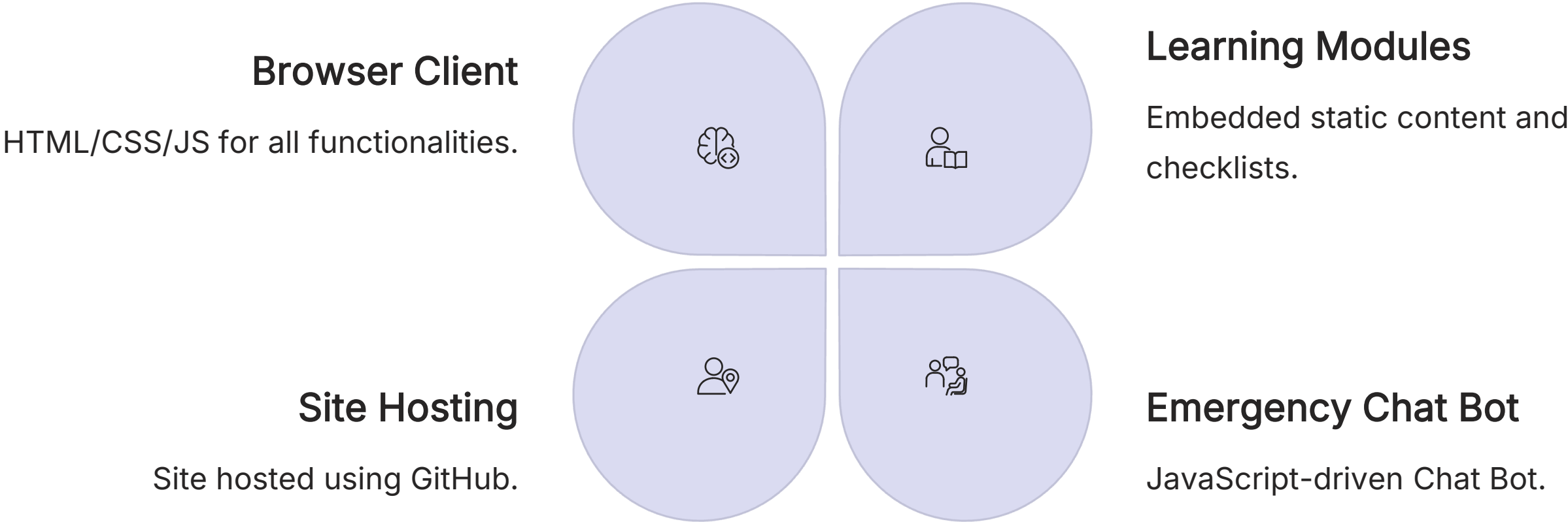
5

Simulated Chat

Use JavaScript logic for interactive guidance.

Methodology: Architecture & Components

The architecture is purely client-side, ensuring maximum accessibility and independence.



Expected Outcomes & Evaluation

Interactive Learning

Users navigate modules and learn procedures with Q&A.

Guidance

Receive advice during emergencies.

Rapid Action

provide clear, quick instructions.

Responsive Design

Adapts to mobile and desktop browsers.

Evaluation focuses on usability, accuracy (adherence to first-aid standards), and performance.

Conclusion: Accessible First-Aid Education

This project delivers a lightweight, offline-capable web application for critical first-aid training and interactive guidance, all without backend servers or external AI.

Its success lies in simplicity, adherence to guidelines, and cross-device usability. Future work could include optional backend integration for analytics or advanced features.

